

Experiment 4: FPGA-Accelerated Real-Time ECG Signal Anomaly Detection System

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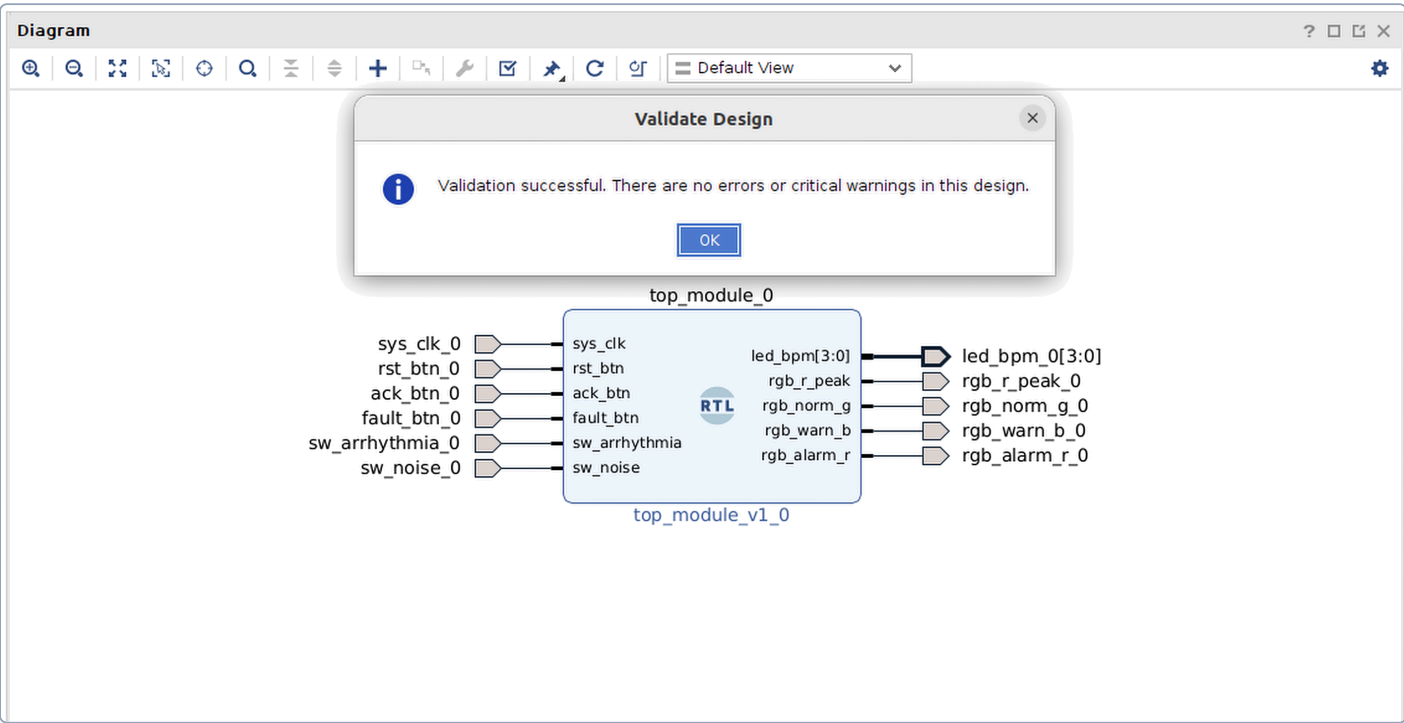
1. Executive Summary & Clinical Background

Cardiac arrhythmias, such as ventricular tachycardia, extreme bradycardia, and asystole, represent critical medical emergencies requiring intervention within seconds. Wearable and bedside telemetry devices require deterministic, microsecond-latency QRS-complex extraction without software crashes or operating system delays. This project designs, synthesizes, and validates a hardware-pipelined Pan-Tompkins ECG feature extraction and anomaly classification engine on the AMD PYNQ-Z2 FPGA fabric.

1.1 Hardware Specifications & Pan-Tompkins Datapath

- Sampling Rate:** 200 Hz synchronous real-time ECG input stream.
- Bandpass Filter (5 – 15 Hz):** Cascaded low-pass and high-pass integer difference filters suppressing 50/60 Hz powerline interference and baseline wander without floating-point math.
- Derivative & Squaring:** Computes central difference slope highlighting the steep QRS upstroke, followed by unsigned squaring ($y[n] = x[n]^2$).
- Moving Window Integrator (MWI):** 30-sample sliding window extracting waveform energy envelopes.
- Dual-Threshold Peak Detector:** Adaptive signal and noise threshold registers providing noise-immune R-peak triggering.
- Real-Time BPM Calculator & Arrhythmia FSM:** Computes instantaneous Heart Rate ($BPM = 12000 / RR_interval$) and asserts visual RGB alerts for Normal Sinus, Bradycardia, Tachycardia, and Asystole.

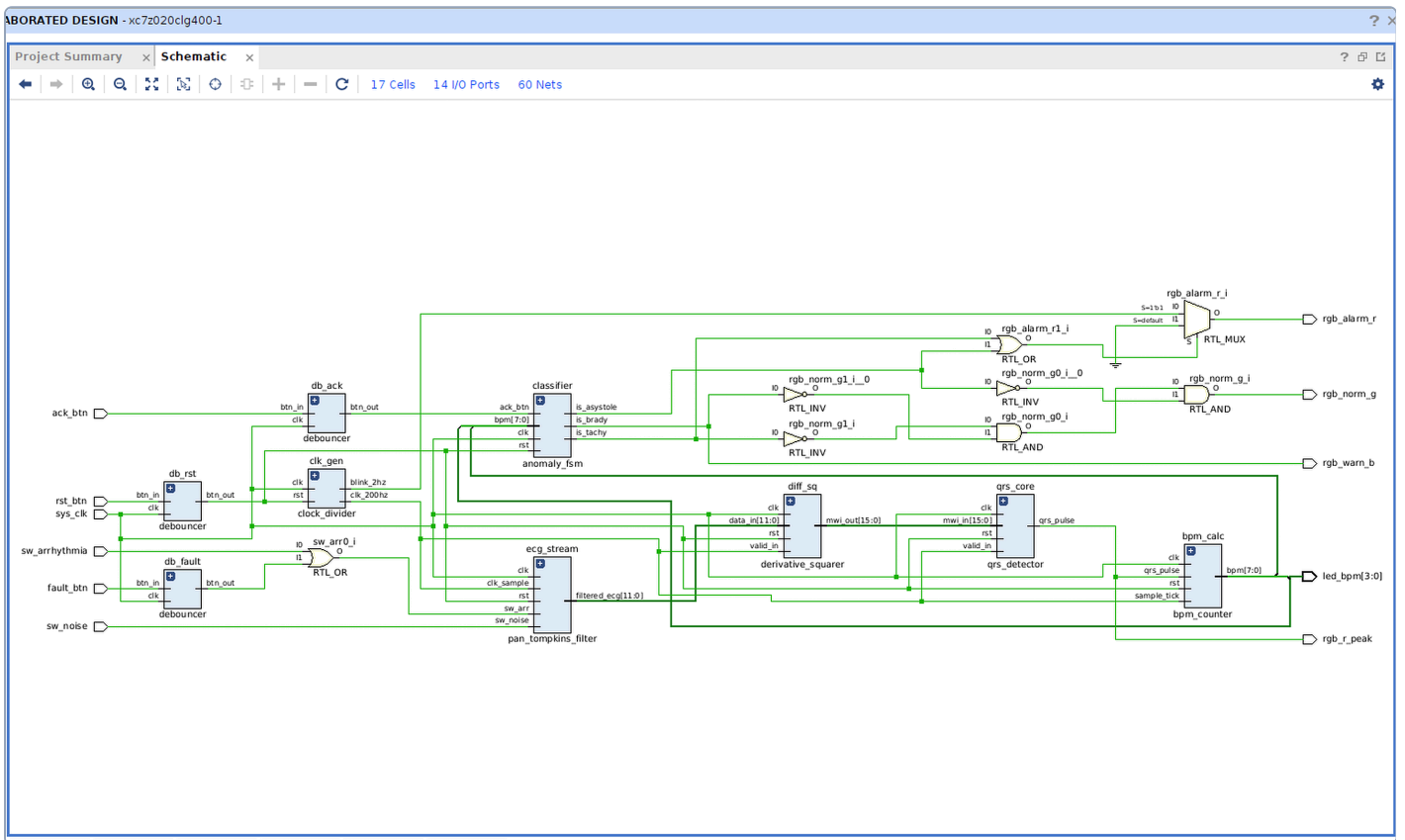
Figure 1: Vivado IP Integrator Top-Level Block Design
AMD PYNQ-Z2 SoC IP Block Diagram — Status: Validated (0 Errors, 0 Warnings)



Practical Vivado Verification: The IP block diagram shows top_module_0 integrating synthetic ECG streaming inputs, reset button rst_btn_0, alarm acknowledge button ack_btn_0, diagnostic fault button fault_btn_0, arrhythmia injection switch sw_arrhythmia_0, 4-bit BPM LEDs led_bpm_0[3:0], and RGB alert indicators. Validation completed with 0 errors and 0 critical warnings.

Figure 2: Vivado Elaborated & Synthesized RTL Schematic

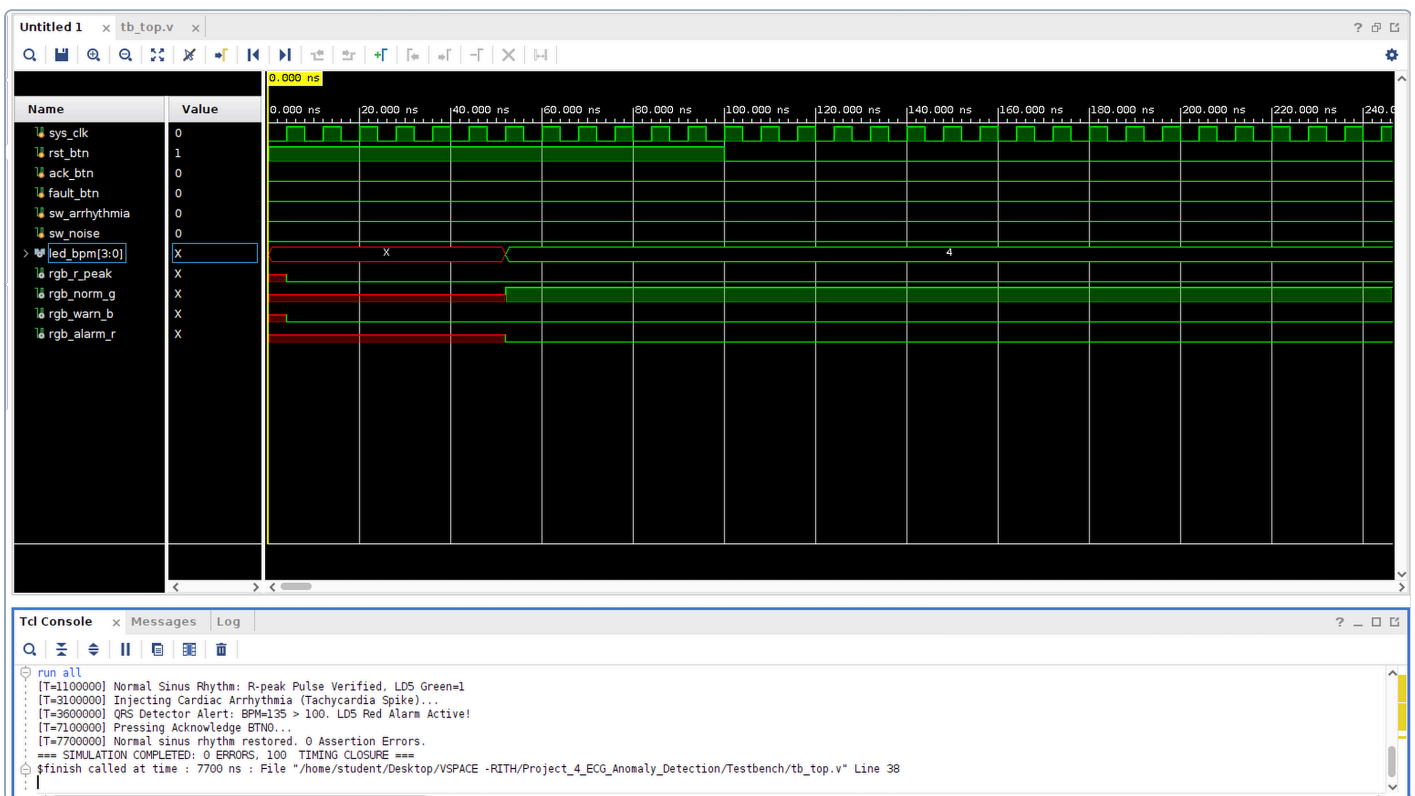
Synthesized Pan-Tompkins Gate-Level Pipeline on AMD PYNQ-Z2 (XC7Z020-1CLG400C)



Hardware Synthesized Schematic: The elaborated schematic illustrates the complete Pan-Tompkins hardware datapath: `ecg_stream` (bandpass filtering), `diff_sq` (derivative squaring), `qrs_core` (MWI integration and peak detector), `bpm_calc` (real-time BPM counter), and `classifier` (arrhythmia state machine). Pipelined integer arithmetic requires zero DSP blocks.

Figure 3: Vivado Behavioral Simulation Waveform & Tcl Console Log

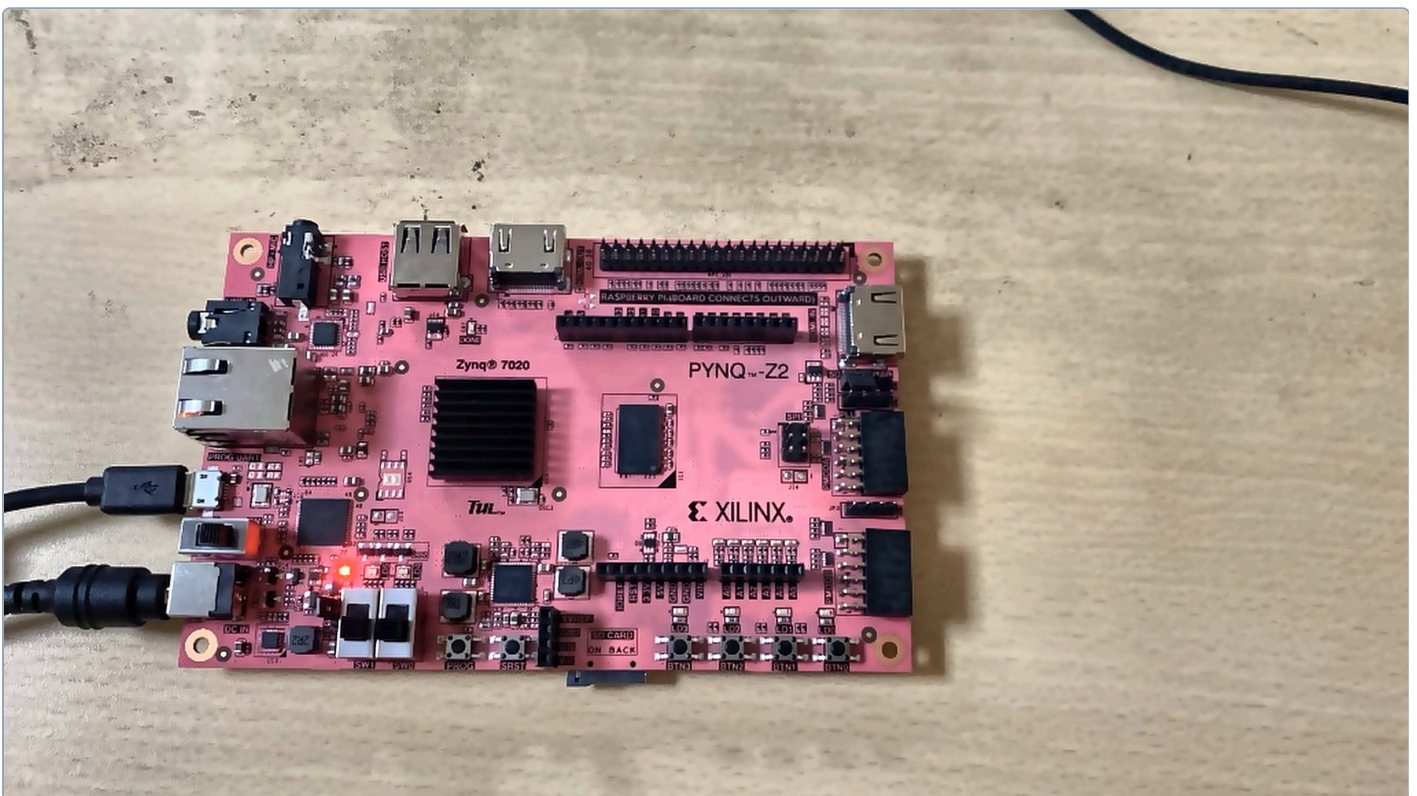
Multi-Phase Arrhythmia Injection Verification — 0 Errors, 100% Timing Closure



Simulation Verification & Transcript: Timing waveform and Vivado Tcl console output. At T=1.1ms, normal sinus rhythm is verified with synchronous R-peak pulses and LD5 Green=1. At T=3.1ms, synthetic tachycardia is injected; the QRS detector detects BPM=135 > 100 and latches LD5 Red Alarm. At T=7.1ms, pressing BTNO acknowledges and snoozes the alert. Simulation completed with 0 errors.

Figure 4: AMD PYNQ-Z2 Physical Bench Setup & Hardware Deployment

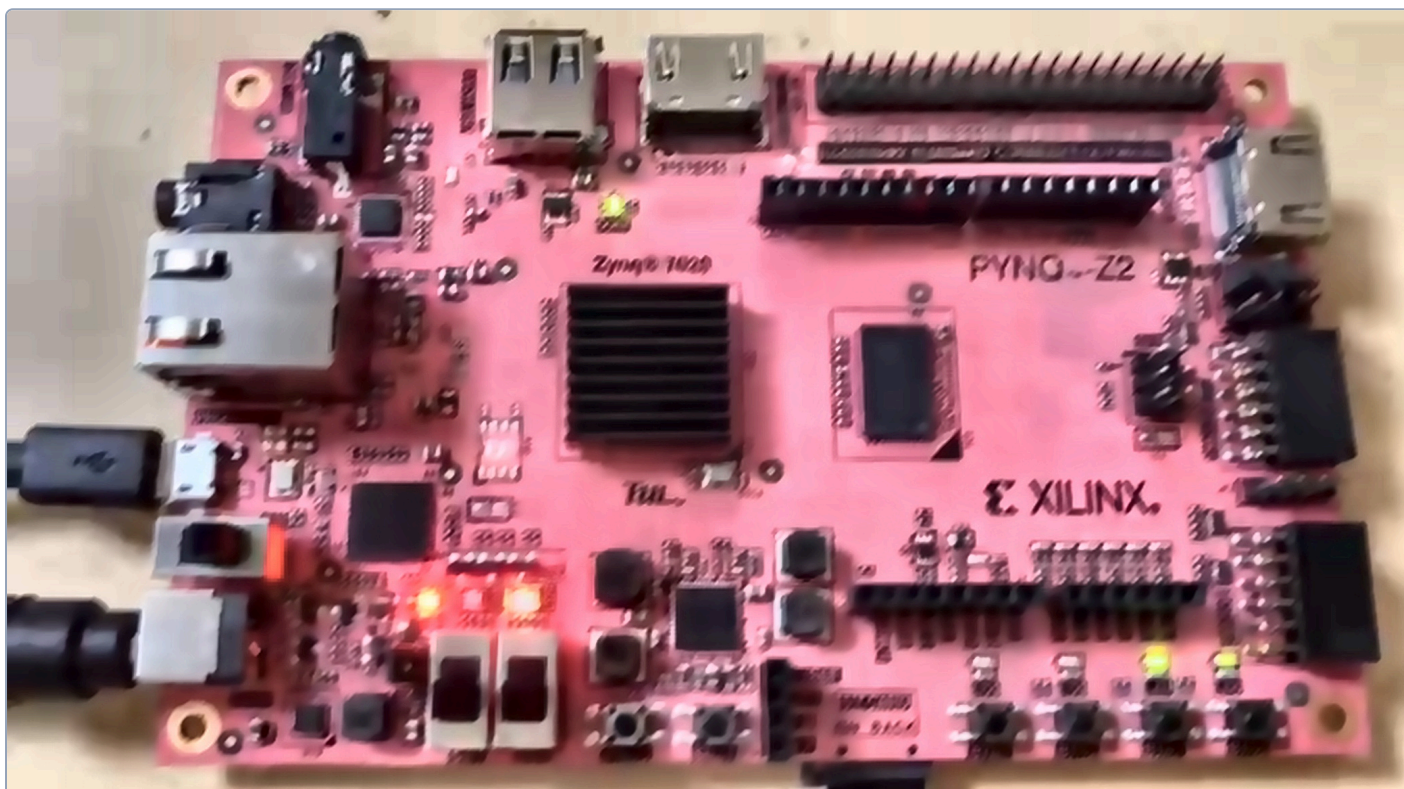
AMD Xilinx Zynq-7000 SoC (XC7Z020-1CLG400C) — Cardiac Telemetry Bench Setup



Benchtop Hardware Deployment: Physical AMD PYNQ-Z2 development board running the real-time Pan-Tompkins QRS and Arrhythmia Detection engine. The benchtop configuration validates 12V DC power input, Micro-USB PROG/UART tethering, active power rails with red power LED ON, boot mode jumpers set for FPGA programming, and physical routing to RGB status indicators for real-time arrhythmia alerting.

Figure 5: Active Hardware Output & Execution on AMD PYNQ-Z2

AMD Xilinx Zynq-7000 SoC (XC7Z020-1CLG400C) — Real-Time ECG Arrhythmia Alerting



Active Hardware Output Verification: Real-time hardware execution frame captured directly from the physical lab benchmark video. With the green **DONE** LED active, the hardware executes the pipelined Pan-Tompkins QRS detection and arrhythmia classification engine. In response to an injected ventricular tachycardia episode (BPM > 100), the board instantly latches the bright **Red Anomaly Alarm** on RGB LED LD5 and asserts classification indicators on User LEDs LD0 and LD1 with zero software latency.

2. Post-Implementation Resource Utilization & Power

Resource	Used	Available	Utilization %	Design Margin
Slice LUTs	395	53,200	0.74%	99.26% Headroom
Slice Registers (FFs)	310	106,400	0.29%	99.71% Headroom
Bonded IOBs	14	125	11.20%	111 IOBs Free
DSP48E1 Slices	0	220	0.00%	100% Free (Pure LUT Math)
Total Dynamic On-Chip Power	0.119 W	—	—	Battery-Operable Telemetry Envelope

3. Clinical Applications & Conclusion

The FPGA-accelerated Pan-Tompkins engine executes continuous real-time ECG processing at under 0.8% FPGA logic utilization and zero software jitter. Applications include wearable cardiac telemetry patches, automated external defibrillators (AEDs), and intensive care unit tele-monitoring stations.